

Challenges Faced

- Spark Environment Setup: Configuring Apache Spark with PySpark, Java, and the required environment variables in WSL2 was initially challenging.
- Dataset Schema Validation: Understanding and validating the automatically inferred data types of different columns required careful checking.
- Revenue Calculation: The calculated revenue using `Quantity_Sold × Unit_Price` differed from the existing `Sales_Amount` column, so both fields had to be understood separately.
- Data Aggregation: Performing category-wise, product-wise, and region-wise analysis using `groupBy()`, `agg()`, `sum()`, and `orderBy()` required proper Spark syntax.
- Spark Execution Logs: Spark generated a large amount of execution information in the terminal, making the actual output sometimes difficult to identify.
- Handling Errors: An error occurred while calculating the average sales amount because an incorrect argument was passed to the aggregation function; it was corrected using `avg()`.
- Dataset Limitation: The provided dataset contained a `Region` column but no `City` column, so the city-wise analysis was adapted to region-wise revenue analysis.